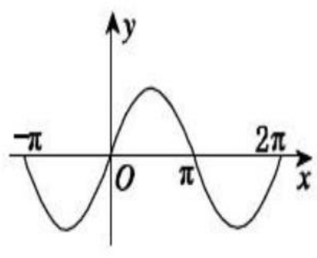
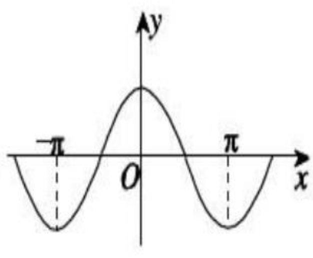
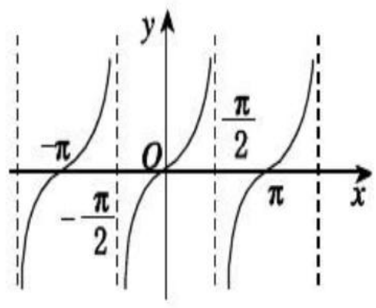
7.6 Inverse Trigonometric Functions

Inverse trigonometric functions arise when we want to calculate angles from side measurements in triangles. They also provide useful antiderivatives and appear frequently in the solutions of differential equations. This section shows how these functions are defined, graphed, and evaluated, how their derivatives are computed, and why they appear as important antiderivatives.

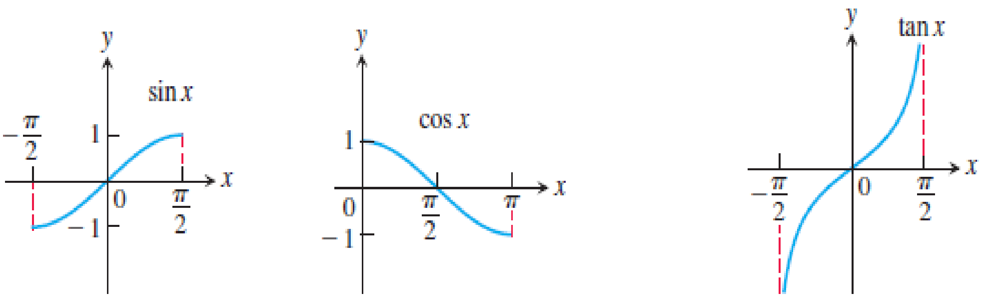
① Defining the Inverse Trigonometric Functions

The six basic trigonometric functions are not one-to-one (since their values repeat periodically).

$y=\sin x$	$y=\cos x$	$y=\tan x$
		
$\mathbf{R}$	$\mathbf{R}$	$\{x \mid x\in\mathbf{R}$, 且 $x\neq k\pi+\frac{\pi}{2}, k\in\mathbf{Z}\}$
$[-1, 1]$	$[-1, 1]$	$\mathbf{R}$

However, we can restrict their domains to intervals on which they are one-to-one.

Domain restrictions that make the trigonometric functions one-to-one



$y = \sin x$ Domain: $[-\pi/2, \pi/2]$ Range: $[-1, 1]$

$y = \cos x$ Domain: $[0, \pi]$ Range: $[-1, 1]$

$y = \tan x$ Domain: $(-\pi/2, \pi/2)$ Range: $(-\infty, \infty)$

Since these restricted functions are now one-to-one, they have inverses, which we denote by

$$y = \sin^{-1} x \text{ or } y = \arcsin x, \quad y = \cos^{-1} x \text{ or } y = \arccos x$$

$$y = \tan^{-1} x \text{ or } y = \arctan x, \quad y = \cot^{-1} x \text{ or } y = \operatorname{arccot} x$$

$$y = \sec^{-1} x \text{ or } y = \operatorname{arcsec} x, \quad y = \csc^{-1} x \text{ or } y = \operatorname{arccsc} x$$

These equations are read "y equals the arcsine of x" or "y equals arcsin x" and so on.

② The Arcsine, Arccosine and Arctangent Functions

DEFINITION

$\boldsymbol{y = \arcsin x}$ is the number in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ for which $\sin y = x$.

$\boldsymbol{y = \arccos x}$ is the number in $[0, \pi]$ for which $\cos y = x$.

$\boldsymbol{y = \arctan x}$ is the number in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which $\tan y = x$.

Example 1 Evaluate (a) $\arcsin\left(\frac{\sqrt{3}}{2}\right)$ and (b) $\arccos\left(-\frac{1}{2}\right)$.

Using the same procedure illustrated in Example 1, we can create the following table of common values for the arcsine and arccosine functions.

x	$\arcsin x$	$\arccos x$
$\sqrt{3}/2$	$\pi/3$	$\pi/6$
$\sqrt{2}/2$	$\pi/4$	$\pi/4$
$1/2$	$\pi/6$	$\pi/3$
$-1/2$	$-\pi/6$	$2\pi/3$
$-\sqrt{2}/2$	$-\pi/4$	$3\pi/4$
$-\sqrt{3}/2$	$-\pi/3$	$5\pi/6$

x	$\arctan x$
$\sqrt{3}$	$\pi/3$
1	$\pi/4$
$\sqrt{3}/3$	$\pi/6$
$-\sqrt{3}/3$	$-\pi/6$
-1	$-\pi/4$
$-\sqrt{3}$	$-\pi/3$

③ Identities Involving Arcsine, Arccosine and Arctangent

$$\sin\left(\sin^{-1} x\right) = x, \quad \cos\left(\cos^{-1} x\right) = x, \quad \tan\left(\tan^{-1} x\right) = x.$$

$$\arccos x + \arccos(-x) = \pi.$$

$$\arcsin x + \arccos x = \frac{\pi}{2}.$$

④ The Derivative of $y = \arcsin u$, $y = \arccos u$ and $y = \arctan u$

Recall **THEOREM 1—The Derivative Rule for Inverses**

If f has an interval I as domain and $f^{-1}(x)$ exists and is never zero on I, then f^{-1} is differentiable at every point in its domain (the range of f). The value of $(f^{-1})'$ at a point b in the domain of f^{-1} is the reciprocal of the value of f' at the point $a = f^{-1}(b)$:

$$\left(f^{-1}\right)'(b) = \frac{1}{f'\left(f^{-1}(b)\right)} = \frac{1}{f'(a)}.$$

We find the derivative of $y = \arcsin x$ by applying Theorem 1 with $f(x) = \sin x$ and $f^{-1}(x) = \arcsin x$:

$$\begin{aligned}\left(f^{-1}\right)'(x) &= \frac{1}{f'\left(f^{-1}(x)\right)} \\ &= \frac{1}{\cos(\arcsin x)}\end{aligned}$$

$$= \frac{1}{\sqrt{1-\sin^2(\arcsin x)}}$$
$$= \frac{1}{\sqrt{1-x^2}}.$$

If u is a differentiable function of x with $|u| < 1$, we apply the Chain Rule to get the general formula

$$\frac{\mathrm{d}}{\mathrm{d}x}(\arcsin u) = \frac{1}{\sqrt{1-u^2}} \frac{\mathrm{d}u}{\mathrm{d}x}.$$

Similarly, we get

$$\frac{\mathrm{d}}{\mathrm{d}x}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}. \qquad \frac{\mathrm{d}}{\mathrm{d}x}(\arctan x) = \frac{1}{1+x^2}.$$
$$\frac{\mathrm{d}}{\mathrm{d}x}(\arccos u) = -\frac{1}{\sqrt{1-u^2}} \frac{\mathrm{d}u}{\mathrm{d}x}. \qquad \frac{\mathrm{d}}{\mathrm{d}x}(\arctan u) = \frac{1}{1+u^2} \frac{\mathrm{d}u}{\mathrm{d}x}.$$

Example 2 Find $\frac{\mathrm{d}}{\mathrm{d}x}\left(\arccos x^3\right)$.

Example 3 Find $\frac{\mathrm{d}}{\mathrm{d}x}\left(\arcsin \frac{3}{x^2}\right)$.

Example 4 Find $\frac{\mathrm{d}}{\mathrm{d}x}\left(\arctan(\ln x)\right)$.

⑤ Integration Formulas

The following formulas hold for any constant $a > 0$.

1. $\int \frac{\mathrm{d}u}{\sqrt{a^2-u^2}} = \arcsin\left(\frac{u}{a}\right) + C,\ u^2 < a^2.$
2. $\int \frac{\mathrm{d}u}{a^2+u^2} = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + C,\ u \in \mathbb{R}.$

Example 5 Evaluate

$$(a)\ \int \frac{\mathrm{d}x}{\sqrt{3-x^2}};$$

$$(b)\ \int \frac{\mathrm{d}x}{\sqrt{4x-x^2}};$$

$$(c)\ \int \frac{\mathrm{d}x}{4x^2+4x+2}.$$